Dietary acid load is associated with lower bone mineral density in men with low intake of dietary calcium

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High dietary acid load (DAL) may be detrimental to bone mineral density (BMD). The objectives of the study were to: 1) evaluate the cross-sectional relation between DAL and BMD; 2) determine whether calcium intake modifies this association. Men (n=1218) and women (n=907) ≥60y were included from the National Health and Nutrition Examination Survey 2005–2008. Nutrient intake from 2–24h recalls was used to calculate net endogenous acid production (NEAP) and potential renal acid load (PRAL) (mEq/d). PRAL was calculated from dietary calcium (PRALdiet) and diet + supplemental calcium (PRALtotal). Tests for linear trend in adjusted mean BMD of the hip and lumbar spine were performed across energy adjusted NEAP and PRAL quartiles. Modification by calcium intake (dietary or total) above or below 800 mg/d was assessed by interaction terms. Overall, mean age was 69 ± 0.3y. Among women, there was no association between NEAP and BMD. PRALdiet was positively associated with proximal femur BMD (p trend=0.04). No associations were observed with PRALtotal at any BMD site (P-range: 0.38–0.82). Among men, no significant associations were observed of BMD with NEAP or PRAL. However, an interaction between PRALdiet and calcium intake was observed with proximal femur BMD (p=0.08). An inverse association between PRALdiet and proximal femur BMD was detected among men <800 mg/d dietary calcium (p=0.02); and no associations ≥800 mg/d (p=0.98). A significant interaction with PRALtotal was not observed. In conclusion, when supplemental calcium is considered, there is no association between DAL and BMD among adults. Men with low dietary calcium showed an inverse relation with PRAL at the proximal femur; in women no interaction was observed. This study highlights the importance of calcium intakes in counteracting the adverse effect of DAL on bone health. Further research should determine the relation between DAL and change in BMD with very low calcium intake.